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Coupled Leaf-like and Kresling patterns for origami-inspired soft continuum robots

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ABSTRACT

Origami and origami-inspired structures are widely utilized in engineering, and origami patterns—as well as their unique mechanical properties—are growing in variety as a result. Here, we create a novel crease pattern derived from Kresling origami by incorporating the rigid Leaf-like origami as a building block within the non-rigid Kresling origami. The resulting coupled Leaf-like and Kresling origami inherits foldability, multistability, and tunable stiffness, while also exhibiting opposite chirality and multiple degrees of freedom (DOFs). Owing to these mechanical properties and its enhanced flexibility compared to the traditional Kresling origami, the coupled origami is well-suited for applications in soft continuum robotics. To demonstrate its practical potential, two types of tendon-driven soft continuum robots—a manipulative robot and a mobile robot are constructed employing the coupled origami as their main body. Experiments validate the robots' ability to perform various tasks and motions, including positioning, crawling, turning, and climbing. This study contributes to the advancement of origami design and broadens its applicability in the development of origami-inspired robotic systems.

1. Introduction

Origami has attracted attention in various fields owing to its versatile kinematic properties, design flexibility, reconfigurability, and multi-functionality. Its unique mechanical properties include multi-stability [1–3], tunable stiffness [4–6], negative Poisson's ratio [7], and anisotropic stiffness [8–9], making origami a valuable tool for various engineering systems. In the field of origami research, a key direction is to explore its unprecedented mechanical properties to integrate origami principles into engineering structures [10–13]. Toward this goal, the design and extension of crease patterns is important [14–17]. For example, the Morph pattern, which lies between the Miura-ori and the eggbox origami, achieves a smooth switch through positive and negative Poisson's ratios by changing the mountain/valley assignment of one of the creases [18,19]. The Leaf-like origami, such as the Leaf-in, skew Leaf-in, and Leaf-out patterns, are also explored by utilizing the Miura-ori as a building block [20,21]. The Yoshimura origami is reconfigured by modifying the vertices to allow highly efficient axial folding [22]. In addition, the tristable property of the Kresling origami has recently been validated, with the unit cell popping out as a new stable state [15,23,24]. The derived structures of the Kresling origami also exhibit improved radial closability and rigid foldability [25,26].

Soft continuum robots are gaining increasing attention due to their intrinsic flexibility, smooth deformation, high adaptability, and tunable stiffness. Origami offers multifunctionality and deployability, opening new avenues for advancing soft continuum robots [27–29]. Nonetheless, the challenges for origami-inspired soft continuum robots arise from integrating actuation mechanisms into origami structures. The actuations are dominated by pneumatic-driven [30–33], magnetic-driven [34,35], smart-material-driven [36–40], and electrical/tendon-driven [27,41–43]. The pneumatic-driven systems excel in delivering localized actuation, while magnetic-driven systems enable remote operation. However, both are hindered by challenges such as bulky devices and spatial constraints. The smart-material-driven actuators, mainly

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utilizing shape memory alloys, can be realized by external conditions such as temperature and stress, but require long cooling response times. In contrast, the electrical/tendon-driven actuators are simple and reduce the distance between the starting point on the tendon and the force application point [44]. These actuation strategies are commonly integrated into the Kresling origami modules for soft continuum robots to realize omnidirectional motions such as bending, torsion, and extension [29,31,41,45–53]. However, due to its coupled compression-torsion property, prior works commonly employ the Kresling origami with opposite chirality to construct origami-inspired robots [29,45–48].

In this paper, we propose a versatile, coupled origami by employing the Leaf-like pattern as building blocks of the Kresling configuration. By leveraging the geometric commonalities between these two diverse types of rigid and non-rigid origami, the geometry and configuration of the coupled origami are proposed. This coupled structure inherits the foldability of the Kresling origami, the multistability of the Leaf-like origami, and the tunable stiffness of the integrated patterns. Moreover, the coupled origami exhibits opposite chirality and, as shown in Fig. 1(a), provides enhanced DOFs and omnidirectional flexibility. Additionally, we create soft continuum robots inspired by the coupled origami, including the manipulative robot and the mobile robot. The former resembles invertebrate limbs, such as an elephant trunk and octopus arms, allowing it to perform tasks within its sizable workspace [32,52,54]. The latter realizes multiple peristaltic motions similar to snails and caterpillars [55–57], as shown in Fig. 1(b). We envision that these works will not only promote the application of origami and origami-inspired structures but also provide new insights for soft continuum robotics research.

Compared with previous studies in origami patterns [3–6,14,18–24] and origami robots [27–32,37–46,49–53,55,58–64], our work offers several contributions: (i) This method of coupling existing origami patterns yields another derivative of Kresling origami, while also achieving attractive mechanical properties. (ii) Among its mechanical properties, the enhanced DOFs are prominent in existing origami patterns, broadening the application potential. (iii) Capable of omnidirectional bending and peristaltic motion, the coupled origami-inspired soft continuum robots also exhibit reduced material dependence and improved fatigue resistance—features that are highly advantageous for origami-based applications.

The paper is organized as follows: In Section 2, Kresling and Leaf-like origamis are introduced, the coupled crease pattern is proposed based on their geometric commonality, and its mechanical properties are analyzed. In Section 3, the manipulative robot is constructed, its performance is evaluated using an equivalent mechanism model, and its bending behavior is demonstrated. In Section 4, the mobile robot is constructed, and its configuration, kinematic principle, and locomotion effects are demonstrated. Finally, the main findings of this work are summarized.

2. Coupled origami with Leaf-like and Kresling patterns

Kresling and Leaf-like patterns are two types of classical origami, both exhibiting intrinsic multistability. In this section, these two origami patterns are introduced, and the coupled origami is proposed by utilizing the Leaf-like pattern as a constitutive unit within the Kresling pattern. This innovative approach results in a novel structure with unique properties and enhanced versatility.

2.1. Geometry of Kresling and Leaf-like origami

The Kresling origami, as a representative pattern of the non-rigid origami type, is widely known for its bistable property and the coupled compression-torsion motion. This structure consists of n repeating arrays of cells with alternating mountain and valley creases, as shown in Fig. 2(a). There are two angles α and β in its crease pattern, which play a critical role in the properties of the Kresling origami, allowing for multiple stable states and tunable stiffness through their modulation [22,65]. The Leaf-out origami is a type of Leaf-like origami

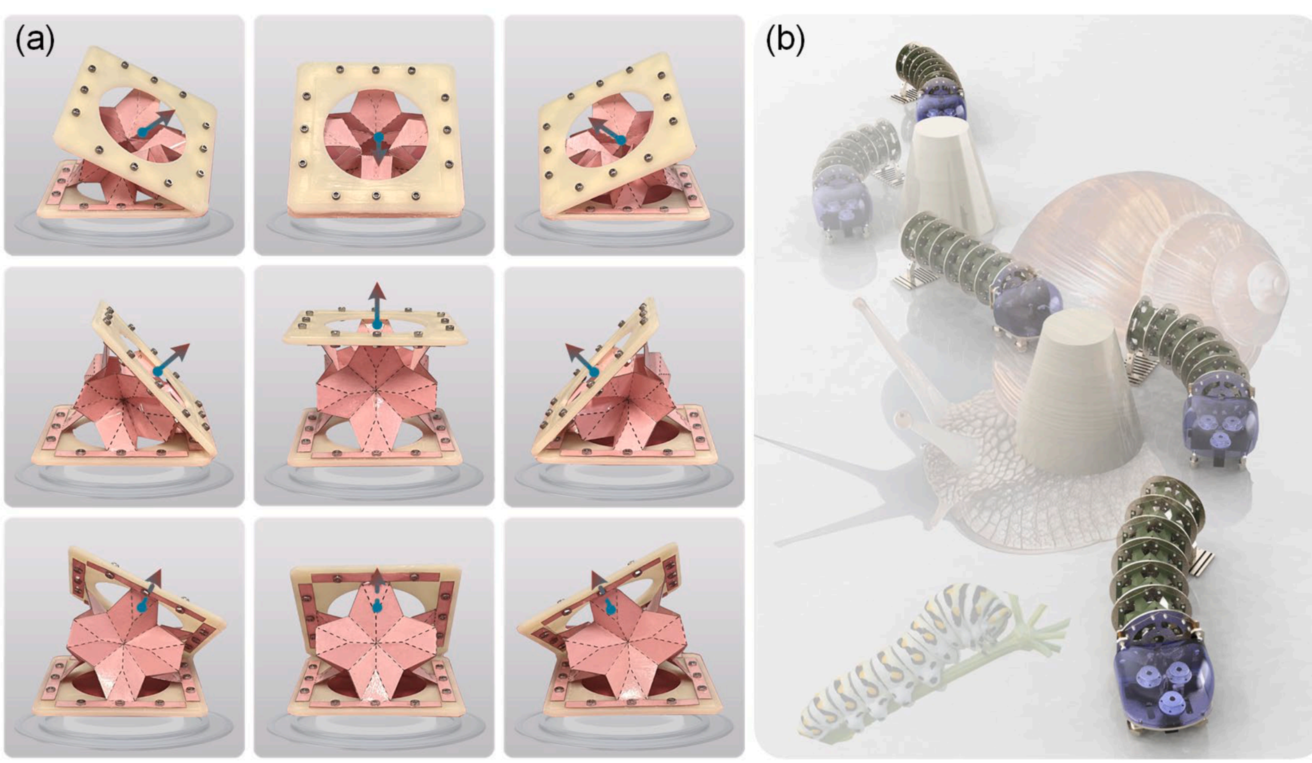

Fig. 1. The coupled origami and its application in a soft continuum mobile robot. (a) Several states of the coupled origami under external force, demonstrating promising flexibility and complex kinematic capabilities. (b) The coupled origami-inspired soft continuum mobile robot demonstrates motions around obstacles, showcasing its potential for versatile locomotion in constrained environments.

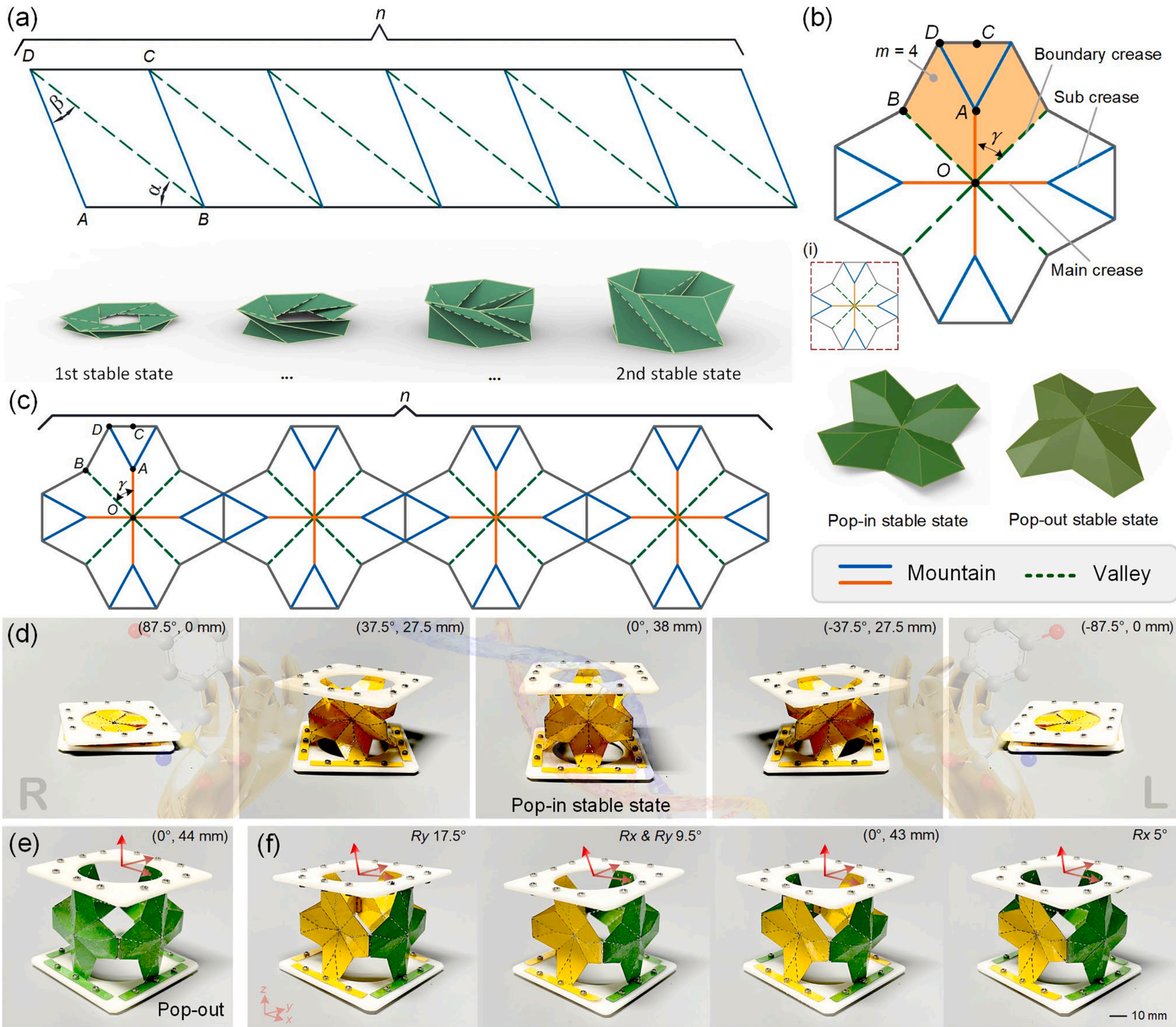

Fig. 2. Geometry of the Kresling origami, the derived Leaf-like origami, and the coupled origami. (a) The crease pattern and folding motion of the Kresling origami. (b) The crease pattern is constructed from the derived Leaf-like module and its two stable states, with its center similar to the Waterbomb origami. (c) The flat sheet with crease patterns consisting of mountain and valley creases of the coupled origami, with the Leaf-like pattern as a building block of the Kresling pattern. (d) The folding motion of the coupled origami can be rotated in both clockwise and counterclockwise directions (opposite chirality). The data in the upper right corner are the rotation angle around the z-axis and the height. (e) The pop-out stable state of the coupled origami is achieved by applying an external force to the pop-in stable state. (f) There are four irregular stable states in the coupled origami as the building blocks gradually pop out. The yellow color indicates the pop-in cell, while the green color indicates the pop-out one. The quantitative data is the result of manual measurements and is directly related to its material.

belonging to rigid origami [20,21], consisting of m repeating unit cells around a single vertex. The main crease, sub-crease, and boundary crease are contained within its unit cell. Here, a derived unit cell of the Leaf-out origami is displayed in the orange part of Fig. 2(b), with its main crease not extended to the outer contour. Interestingly, the multi-module pattern formed by four such derived unit cells features an eight-crease around a single center vertex, resembling the Waterbomb origami with the pop-in and pop-out stable states. Naturally, this pattern also possesses two stable states and undergoes rigid folding transitions between them, as shown in Fig. 2(b). More details for the Kresling and Leaf-like origami are given in *SI Note* 1 and Fig. S1.

2.2. Geometry of coupled origami

Previous research focused on analyzing the geometric parameters α and β of the Kresling origami to achieve various properties, such as multistability, tunable stiffness, and versatile kinematic modes. It is worth noting that the unit cells of the Kresling origami are quadrilaterals, while the outer contour of this derived Leaf-like pattern is also a quadrilateral, as shown in the dark red dotted line in inset (i) of Fig. 2(b). This opens up the possibility of treating the Leaf-like origami as a side wall of the Kresling origami for arranging into a novel crease pattern. Based on this geometric commonality, the coupled origami is proposed, as shown in Fig. 2(c) ($n = 4$). It can be characterized by six parameters, namely the number of units n, the central angle γ (which is the angle between the boundary crease OB and the main crease OA), and four length parameters, L_{OA}, L_{OB}, L_{AC}, and L_{AD}, which are defined based on the geometric definitions of the Kresling origami and the foliated origami. By folding the crease pattern along the predefined mountain and valley creases, a three-dimensional structure resembling Kresling origami is generated, and its folding motion is shown in Fig. 2(d) and Movie *S1*. This coupled origami is categorized as a Kresling-derived structure due to its similar non-rigid folding behavior. Specifically, although each derived Leaf-like unit cell undergoes a rigid transition between the pop-in and pop-out states, the assembly of multiple unit cells introduces additional geometric constraints. These inter-unit connections fundamentally alter the folding mechanism, such that the coupled origami deviates from the rigid origami assumption. Notably, this novel origami could be folded in both clockwise and counter-clockwise directions, implying opposite chirality (left-handedness and right-handedness). Although the concept of chirality has been frequently mentioned in previous research on Kresling origami, it can only be folded in one direction. Here, the coupled origami realizes the motion with opposite chirality in a single module, which is extremely attractive. In essence, the boundary creases in the Leaf-like origami correspond to the valley creases that play the primary role in the chirality of the Kresling origami, and there are two such mirror-symmetric boundary creases in the unit cell of the coupled origami. These two creases, which are tilted to the left and the right, allow the coupled origami to achieve folding motion with opposite chirality. More details for the coupled origami are given in *SI Note* 2 and Fig. S2.

The Leaf-like origami inherently exhibits the pop-in and pop-out stable states [20,21,66,67], and serves as a building block of the coupled origami, indicating that the coupled origami at least holds these two stable states. However, the folding motion of the coupled origami only exhibits the pop-in state, as shown in Fig. 2(d). This is attributed to the Leaf-like and also the Waterbomb patterns, which are unenclosed structures whose stable states require an external, out-of-plane force to switch, unlike the stable state of the multi-triangles cylindrical origami that pops out during the folding process [15]. Therefore, to reach the pop-out stable state, the coupled origami also requires an external force applied at the center of each unit cell, as shown in Fig. 2(e). By enforcing this pop-out state on select side walls, multiple irregular stable states are achieved. Taking the coupled origami with $n = 4$ as an example, there are four distinct states corresponding to one, two, and three units popping out, as shown in Fig. 2(f) and Movie *S1*.

Overall, the rigid Leaf-like pattern and the non-rigid Kresling pattern are coupled to introduce a novel crease pattern that exhibits numerous attractive properties. The multistability of this coupled origami originates from the Leaf-like/Waterbomb pattern, which is manifested as the pop-in and pop-out stable states. The distinct structures of these two states result in varying stiffnesses, which serve as a source of the tunable stiffness in the coupled origami, as shown in *SI Note* 3 and Fig. S3. In contrast, the tunable stiffness of the coupled origami could also be achieved by adjusting the geometric parameters (Fig. S4), which is a property of the Kresling origami. Moreover, the foldability of the coupled origami is closely associated with the Kresling origami, adhering to the geometric condition, $\gamma = \pi / n$, as shown in *SI Note* 4 and Fig. S5 [68]. In general, the coupled origami integrates two distinct types of origami patterns, inheriting properties from both while also introducing unique properties, such as the opposite chirality and multiple DOFs.

3. Coupled origami-inspired soft continuum manipulative robot

The coupled origami shares many properties with the Kresling origami, but one of the most important aspects is that the coupled origami provides enhanced DOFs, demonstrating improved flexibility. From this perspective, the coupled origami provides a versatile platform for soft continuum robots. To prove this, in this section, these enhanced DOFs are analyzed, and the corresponding reachable workspace is obtained using an equivalent parallel mechanism model. In addition, the coupled origami-inspired soft continuum manipulative robot (COISCMR-1) is created, and its ability to perform tasks is displayed.

3.1. Mechanical equivalent model of coupled origami

It is well known that although Kresling origami achieves omnidirectional bending and torsion, these motions rely entirely on material flexibility [15,31,50,65], as shown in *SI Note* 5 and Fig. S6. More seriously, its bending motion is also affected by the coupled torsion and is not independent. However, the coupled origami possesses two rotational DOFs (*Rx* and *Ry*), one axial telescopic DOF (*Tz*), and one coupled compression-torsion DOF (*Rz* & *Tz*), totaling four DOFs when the geometric condition of the foldable property ($\gamma = \pi / n$) is satisfied, as shown in Fig. 3(a). Conversely, the coupled one is lost while the folding condition is not satisfied. In general, the coupled origami possesses at least three DOFs (see Movie *S2*), which enables a sizable workspace and gives more options for actuation methods. To analyze the coupled origami, it can be treated as a spatial $P^{*1}SP^{*2}$ parallel mechanism with the same DOFs, where the 'P^{1}'and 'P^{2}' stand for prismatic joints, the 'S ' for spherical joint, and the '*' indicates that these two prismatic joints are required to move symmetrically, as shown in Fig. 3(b). This equivalent mechanism is driven by four flexible ropes, with the two prismatic joints moving synchronously to maintain symmetry consistent with the coupled origami (similar to the *PS* mechanism [69]). By utilizing the inverse kinematics model of the equivalent mechanism and referring to the boundary conditions of the coupled origami (as shown in *SI Note* 6 and Fig. S7), the reachable positions of the moving platform are obtained, as shown in Fig. 3(c). It is worth noting that the three individual DOFs (*Rx, Ry,* and *Tz*) in the coupled origami facilitate its control strategy, avoiding the twisting effects commonly suffered in the Kresling origami and allowing for direct actuation through tendons.

3.2. Functionality of the soft continuum manipulative robot

We connect five origami modules in series as the main body to create the COISCMR-1 driven by four tendons (more details are shown in *SI Note* 7 and Fig. S8), aiming to achieve a larger workspace and excellent ability to perform tasks by exploiting the flexibility of the coupled origami, as shown in Fig. 3(d). The origami modules primarily govern the deformation path and movement behavior of the COISCMR-1, but they are unable to recover due to their limited stiffness. To facilitate and stabilize the recovery process, springs are incorporated along the tendons between modules as auxiliary elements (bellows acting as guides for the springs, as shown in Fig. S9). Through separate control between the tendons, the COISCMR-1 achieves different bending degrees and axial compression, as shown in Fig. 3(e) and Movie *S3*. The COISCMR-1 focuses only on the axial DOF (*Tz*) of the coupled origami, rather than the coupled one (*Tz* & *Rz*), resulting in a partially compressed state under this straightforward actuation (the compressibility is about 64.10 %), which is also constrained by the springs and bellows. However, utilizing the coupled DOF to achieve a compact compressed state similar to the Kresling origami is also structurally permissible, as shown in Fig. S8. The COISCMR-1 enables omnidirectional bending motions only using a simple control between the tendons, as shown in Fig. 3(f) and Movie *S4*. For instance, 45° oblique bending is achieved by applying the same actuation displacement to the tendons 1 and 2. For axial bending along the *x*-axis (0°), an actuation displacement is applied to the tendon 1, while another relatively smaller displacement is applied to the tendons 2 and 4. For compression and extension of the COISCMR-1, the same displacement is applied to the four tendons. Its attitude allows it to become more varied by adjusting the control strategy.

The COISCMR-1 is considered a series of the equivalent mechanisms, and as the stiffness of each module is uniform, the relationship between this robot's end and the tendons is established to predict its different postures. The 300 randomized working paths of the COISCMR-1 are obtained, and these flexible postures allow it to reach positions that form a sizable workspace, indicated by the blue dots, as shown in Fig. 3(g). These working paths and reachable positions are gained by considering the springs and bellows in the COISCMR-1. Due precisely to these components, the rotation around the *Z*-axis (*Tz & Rz*) in the coupled origami is limited, and the motion range of the remaining three DOFs is also somewhat constrained, resulting in a slightly smaller than ideal

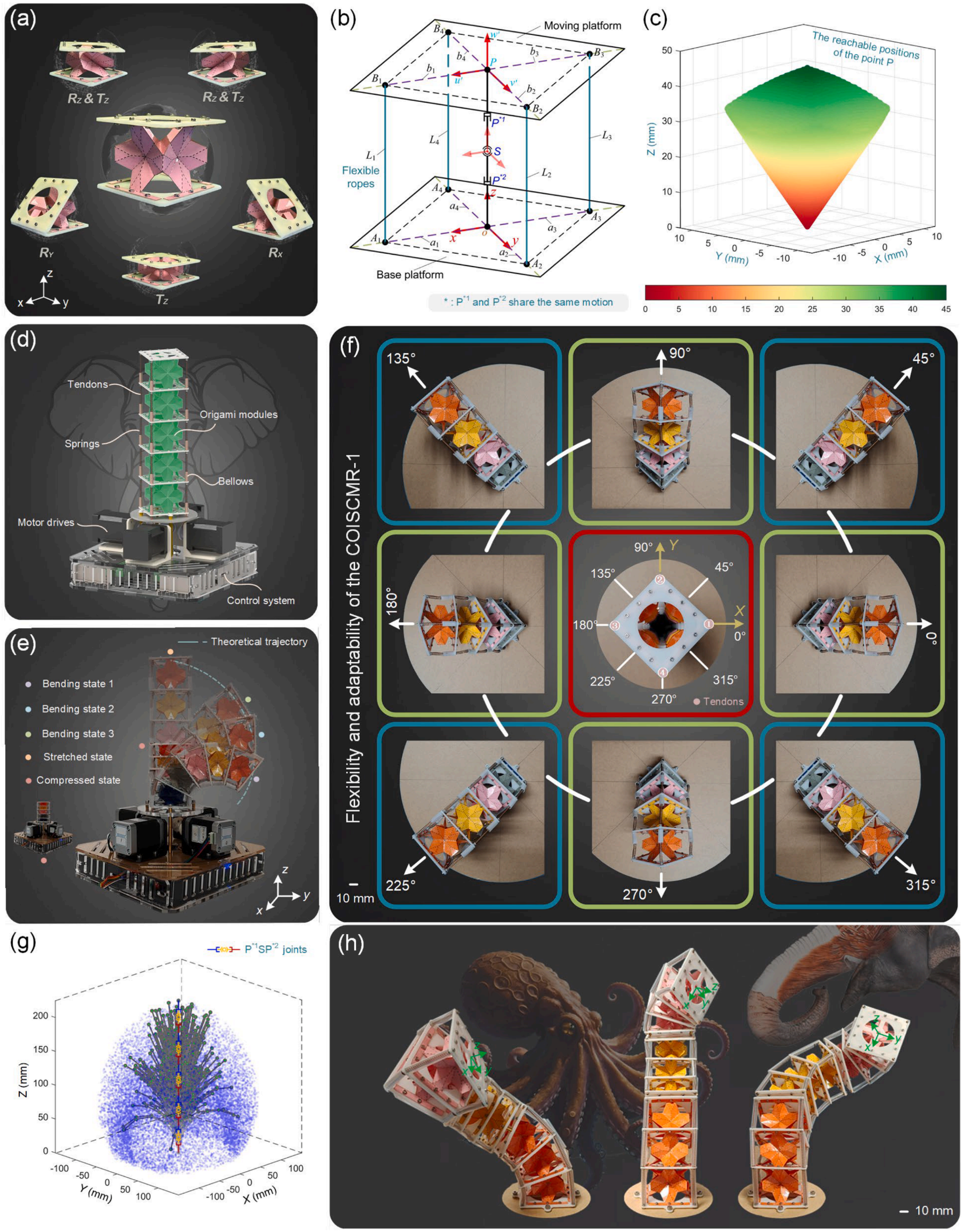

Fig. 3. The coupled origami-inspired soft continuum manipulative robot. (a) There are four DOFs in the coupled origami with foldable property. (b) The equivalent mechanism of the coupled origami is driven by four flexible ropes. (c) The reachable positions of the equivalent mechanism, with symmetry consistent with the coupled origami. (d) In the prototype of the COISCMR-1, the main components include five origami modules, four tendons and driving motors for transmitting forces, the springs for quick recovery, and a set of control systems. (e) The stretched state, the compressed state, and three different bending states demonstrate the ability to perform tasks as a manipulative robot. (f) The omnidirectional bending motions of the COISCMR-1 demonstrate excellent flexibility and provide a versatile platform for soft continuum robots. (g) The 300 randomly generated working paths and the workspace of the COISCMR-1. The former is indicated by 5 line segments ($P^{*1}SP^{*2}$ joints) in series with 6 endpoints (5 origami modules), and the latter is composed of blue dots. (h) The three states of the origami robotic arm contain three segments with different drives/stiffnesses, each segment consisting of three origami modules, demonstrating its potential to perform complex tasks.

workspace, see *SI Note* 7 for more details. Whether analyzed as continuum models or equivalent mechanisms [70–72], the control and operation of the COISCMR-1 is extremely convenient, owing to the flexibility and adaptability guaranteed by the multiple DOFs of the coupled origami. To further demonstrate the potential of the coupled origami in enabling complex motions, we assemble a three-segment continuum configuration along the lines of the COISCMR-1. In this prototype, each segment is actuated independently using tendons with different input parameters, thereby inducing various stiffness and curvature across the segments. This strategy allows each segment to adopt a distinct posture, thereby enhancing its adaptability and its ability to perform complex tasks, as shown in Fig. 3(h). Assembling multi-segment configurations via complex actuation systems is a common practice in soft continuum robots [40,72–80]. The demonstration in Fig. 3(h) evidences that the coupled origami enables independently controlled segmental deformation, highlighting its potential as an option for applications in soft continuum robotics. In addition, it is worth noting that this robotic system only exploits the pop-in state of the coupled origami, as the pop-out state has a higher stiffness and is therefore not conducive to deformation, as discussed in *SI Note 4*. However, the tunable stiffness of the coupled origami provides options for the COISCMR-1, and the capability to enhance stiffness in the pop-out state is explored in *SI Note 8* and Fig. S10.

Due to the limited stiffness of the origami module and the current actuation method, the COISCMR-1 employs springs and bellows to assist its functional performance. However, the deformation trajectory, directional compliance, and overall shape-shifting behavior of the COISCMR-1 are primarily governed by the geometric configuration and mechanical properties of the coupled origami module. Previous studies have also confirmed that such origami-based architectures exhibit excellent structural response capabilities under magnetic or pneumatic actuation [16,33,34,50,51]. Therefore, with a more suitable actuation scheme, the COISCMR-1 is expected to further unleash the potential of origami modules and achieve more pronounced display effects and performance. Compared with the previous Kresling origami robots that rely on the material flexibility to achieve bending effects [15,16,33,51], the COISCMR-1 achieves varying degrees of bending and omnidirectional bending enabled by the multiple DOFs of the coupled origami and allowed by its structural mechanism. Origami tower-inspired robots [52,53] alleviate the material dependence to some extent, but their unit cells require multiple components, which raises challenges to the manufacturing approach. Meanwhile, Kresling origami and origami tower-inspired robots also suffer from a non-negligible disadvantage in that their extensibility is inevitably affected by their torsion, rendering tendon actuation infeasible or necessitating the stacking of two-layered structures with opposing chirality to mitigate this effect. In addition, origami-inspired structures applied to manipulation robots [41,49] require consideration of stiff-flexible coupling properties within the material to mimic the flexibility of origami, potentially leading to the loss of some inherent origami properties. In contrast, we achieve both flexibility and controllability by leveraging the multiple DOFs in the coupled origami, without being hindered by its chirality and material deformation. The efficacy of the proposed system is further elaborated on in the following section.

4. Coupled origami-inspired soft continuum mobile robot

Another promising application of the coupled origami lies in mobile robotics, where it offers the advantage of an axial DOF that can operate independently, rather than being coupled with its torsion. Therefore, the coupled origami-inspired soft continuum mobile robot (COISCMR-2) is created with the coupled origami as its main body. In this section, its working principle and kinematic model are discussed, and its ability to move continuously in complex obstacle environments is showcased through experiments.

4.1. Principle of the soft continuum mobile robot

Utilizing the free-standing extensibility of the coupled origami, we create the COISCMR-2, which consists of six origami modules interconnected by three tendons, as shown in Fig. 4(a). These three tendons (T1, T2, and T3) are evenly distributed and actuated by three motors, as shown in the front view inset. The origami modules can be compressed by pulling the tendons and then be released, relying on their own elasticity and the restoring force of the built-in springs. The springs in the COISCMR-2 function in the same way as those in the COISCMR-1—they merely assist in the recovery of the origami modules and are not the primary actuation source. This cycle of pulling and releasing enables the COISCMR-2 to move forward. Additionally, the synergy between friction and gravity is also crucial for its locomotion. While the friction plate is in the state 1, as shown in inset (i) of Fig. 4(a), it appears tilted and loses its frictional effect with the ground, allowing it to be dragged forward when the tendons are pulling. By contrast, in the state 0, as shown in inset (ii) of Fig. 4(a), the friction plate maintains contact with the ground, delivering friction to propel the head forward when the tendons are released. The remaining components, such as the drive motors, control system, and battery, are located at the front so that the center of gravity remains over the head. The purpose is to ensure that the head remains stationary while driving the tendons, thus, it relies on the folding of the origami modules to drag the friction plate forward. This also prevents the COISCMR-2 from falling backward when the tail is tilted (in the state 1). More details for the COISCMR-2 are given in *SI Note* 9.

To analyze its underlying principle, we divide the crawling motion of the COISCMR-2 into distinct phases, as shown in Fig. 4(b). In the first phase, only the tendon T1 is actuated to adjust the friction plate to the state 1. In the second phase, all three tendons are actuated at the same time, compressing the origami modules and dragging the friction plate forward. Then, only the tendon T1 is reversely actuated to release the displacement applied in the first phase, aiming to restore the friction plate to the state 0, as shown in the third phase. In the fourth phase, the three tendons are again actuated in the opposite direction simultaneously, returning the origami modules to their initial state and pushing the head forward. With the four phases as a cycle, the COISCMR-2 achieves forward movement along with the compression and recovery of the origami modules. The climbing function operates on the same principle but is affected by gravity, resulting in different behaviors for uphill and downhill motions—the latter typically results in greater displacement. In contrast, the turning function involves slightly more complex mechanisms (*SI Note* 9), but only requires controlling the difference between the tendons T2 and T3 (different drive rates in these two motors). The core of this turning process is to control the bending of the origami modules through the difference in the tendons, to adjust the friction plate for turning first, and then rely on lateral friction to achieve the rotation of the head in the same direction as the friction plate, as shown in Fig. 4(c) and Fig. S11. Under the principle of center of gravity and friction synergy, the performance of the COISCMR-2 may be influenced by various factors, including its weight, the elasticity of the origami modules, the stiffness of the spring, and others, as detailed in *SI Note* 10.

4.2. Kinematic model of the soft continuum mobile robot

The kinematic principle indicates that while the tendons (T2 and T3) are controlled simultaneously, the origami modules push the COISCMR-2 to wriggle forward. Conversely, when these two tendons are controlled differently, the COISCMR-2 achieves turning through the bending of the origami modules. To ensure more accurate movements, the kinematic model of the COISCMR-2 is established, as shown in Fig. 4(d). The assumptions of this model include that the curvature in each module is uniform, and that the errors caused by the manual manufacturing method are ignored. Therefore, the origami modules shown in Fig. 4(d) (i) are closely denoted as the virtual backbone of the continuum

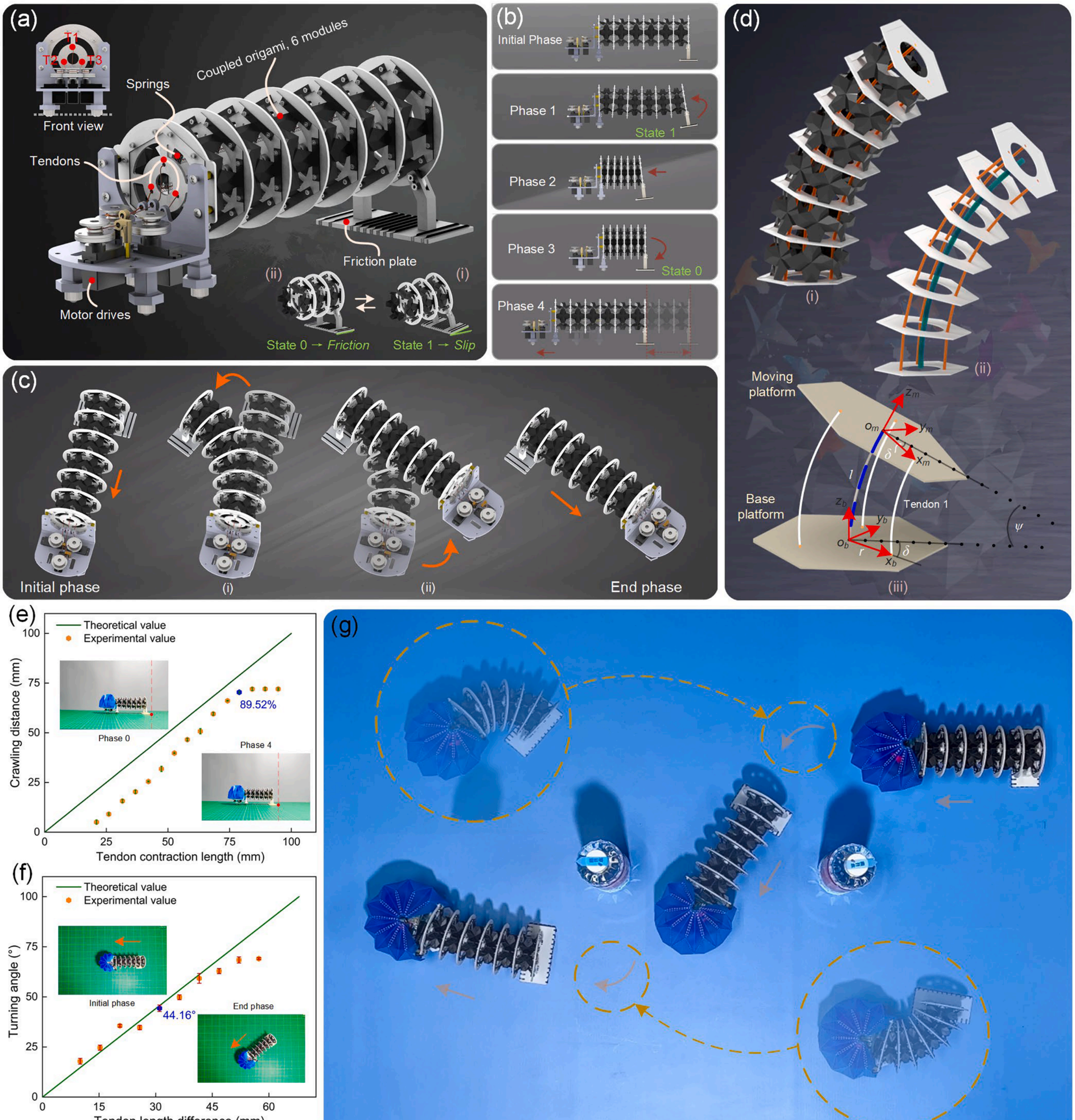

Fig. 4. The prototype, principles, kinematic model, and experiments of the COISCMR-2. (a) In the prototype of the COISCMR-2, the main components include six origami modules, the friction plate that enables forward motion, three tendons and driving motors, the springs for quick recovery, and a set of control systems. (b-c) The crawling and turning motions of the COISCMR-2 are realized by adjusting the friction plate and controlling the drive rate between the tendons. (b) The crawling motion, where one cycle consists of four motion steps. (c) The turning motion relies on different rates of the three tendons. (d) The kinematic model provides important support for guiding the motion of the COISCMR-2. (i) The bending state, including six identical origami modules. (ii) The simplified model of the continuum mechanisms applied to the COISCMR-2. (iii) The kinematic model of the single origami model, with base and moving platforms to express the end pose of the COISCMR-2. (e) Results of experiments and models related to the relationship between the tendon contraction and the crawling distance. (f) Results of experiments and models related to the relationship between the tendon differentials and the turning angles. The data in subplots (e) and (f) are plotted as mean and standard deviation based on five samples. (g) The continuous motion capability of the COISCMR-2 in complex environments with obstacles.

mechanism, as shown in Fig. 4(d)(ii). Taking one module as an example, it is denoted by the corresponding equivalent linkage mechanism, where both ends are referred to as the base and moving platforms, as shown in Fig. 4(d)(iii). By connecting all equivalent mechanisms in series, the motion of the COISCMR-2 can be constructed [27,69,81].

The base coordinate o_b-$x_b y_b z_b$ and the moving coordinate o_m-$x_m y_m z_m$ are located at the center of the base and the moving platforms, respectively. The x-axis points to the tendon 1, and the z-axis is perpendicular to the platform. For a single module, the driving space $\boldsymbol{L}$ can be expressed as $\boldsymbol{L} = [L_1, L_2, L_3]^{\mathrm{T}}$, representing the lengths of tendons under actuation. The configuration space can be defined by the length of the virtual backbone l, the bending angle ψ, and the angle of the bending plane δ. Therefore, the position and rotation matrix of the moving platform relative to the base one can be expressed by

$$\mathbf{p}_b^m = \frac{l}{\psi}[\,(1-cos\psi)cos\delta \quad (1-cos\psi)\sin\delta \quad \sin\psi\,]^{\mathrm{T}}, \tag{1}$$

$$\mathbf{R}_b^m = \mathrm{Rot}(\mathbf{Z}_\mathrm{b},\delta)\mathrm{Rot}(\mathbf{Y}_\mathrm{b},\psi)\mathrm{Rot}(\mathbf{Z}_\mathrm{m},-\delta) = \begin{bmatrix} \cos\psi cos^2\delta + sin^2\delta & (cos\psi-1)cos\delta\sin\delta & cos\delta sin\psi \\ (cos\psi-1)cos\delta\sin\delta & cos^2\delta+\cos\psi sin^2\delta & \sin\delta sin\psi \\ -cos\delta sin\psi & -\sin\delta sin\psi & \cos\psi \end{bmatrix}. \tag{2}$$

The relationship between the configuration parameters and driving space can be derived as follows:

$$\psi = \frac{2\sqrt{L_1^2+L_2^2+L_3^2-L_1L_2-L_2L_3-L_1L_3}}{3r}, \tag{3}$$

$$\delta = \arctan\left(\frac{L_2+L_3-2L_1}{\sqrt{3}(L_2-L_3)}\right). \tag{4}$$

Then, the relationship between the driving space and the configuration parameters is as follows:

$$\begin{bmatrix} L_1 \\ L_2 \\ L_3 \end{bmatrix} = \begin{bmatrix} l + r\psi\cos\delta \\ l - r\psi sin(\pi/6-\delta) \\ l - r\psi\cos(\pi/3-\delta) \end{bmatrix}. \tag{5}$$

The crawling distance of the COISCMR-2, which depends on the controlled tendon contraction length, can be easily predicted, as shown in Fig. 4(e). However, due to the influence of sliding on the friction plate, the elasticity of the tendons, and the origami material, there is a difference between the crawling distance and the tendon contraction length during the experiments. The results indicate that the highest crawling efficiency is about 89.52 %. The turning principle shown in inset (i) of Fig. 4(c) reveals that the bending plane of the end platform coincides with the *yoz* plane, which means that $\delta = 90^\circ$ and $L_2 + L_3 = 2L_1$. This indicates that the difference between the tendons T2 and T3 realizes bending, while the tendon T1 is passively actuated, confirming the turning principle of the COISCMR-2. Then, the relationship between the tendon length difference and the bending angle is also obtained, as shown in Fig. 4(f). Another important factor contributing to the error in the turning tests is the lateral slip that occurs on the friction plate, which causes the experimental data to fluctuate around the theoretical value, and the maximum turning angle achievable is 69.03° The performance parameters of the COISCMR-2 are shown in Table 1.

4.3. Mobility of the soft continuum mobile robot

We conduct experiments to evaluate the locomotion performance of the COISCMR-2 on a rubber board (Fig. S12). Two cameras are utilized to record the front and top views. The times of each sub-step during crawling were 1.5 s, 6.0 s, 0.5 s, and 13.5 s. The first two of these are the times at which the tendons are pulled to compress the origami modules, and the last two are the times when the tendons are released. The initial length of the six origami modules is 144 mm, which is reduced to 65 mm in the phase 3, resulting in a compression ratio of 54.9 %. The crawling effect of the COISCMR-2 for a single cycle is shown in Movie *S5*, with a measured forward movement of 70.63 ± 0.56 mm, corresponding to contracting the tendons by 78.9 mm. Then, the climbing tests are carried out in an environment with a 5° slope to assess the ability to move both uphill and downhill. The results using a single cycle indicate that the COISCMR-2 moves 81.22 ± 0.64 mm forward on a downhill slope and only 49.30 ± 0.69 mm on an uphill slope, as shown in Movie *S6*. This result is related to the design of the COISCMR-2, where its center of gravity causes it to be dragged forward by inertia and gravity on a downhill slope, resulting in a longer moving distance than the crawling distance in a flat environment. Conversely, the COISCMR-2 is less effective on an uphill slope. For the turning tests of the COISCMR-2, the times for each sub-step are 3.0 s, 1.2 s, 5.5 s, and 14.3 s, as shown in Movie *S7*. While adjusting the state of the friction plate, the head slips slightly sideways due to the poor lateral friction. During the turning process of the head shown in inset (ii) of Fig. 4(c), the friction plate also slips sideways. However, these phenomena have always existed and are consistent with the principles of the COISCMR-2 that rely on friction to achieve turning, which ultimately achieves a turning angle of approximately 44.16 ± 1.46° (the theoretical value is 45.65°). For this lateral friction, another friction scheme is also discussed, as shown in *SI Note* 10 and Fig. S13. These experiments validate the kinematic principle underlying the COISCMR-2 and demonstrate its expected performance, showcasing the potential of the coupled origami in mobile robots. Furthermore, the continuous locomotion of the COISCMR-2 is also displayed through the integration of a remote control handle, and its ability to move around obstacles is shown in Fig. 4(g) and Movie *S8*. Its robustness and stability are also evaluated, as shown in *SI Note* 10 and Fig. S14. The performance analysis compared to previous origami mobile robots [29,30,42,45–47,55,60–64] is given in *SI Note* 11.

Table 1
Relevant performance parameters of the COISCMR-2.

Evaluation indicators	Performance parameters
Total length	269.10 mm
Total weight	290.84 g
Average crawling speed	3.29 mm/s
Maximum crawling efficiency	89.52 %
Maximum turning angle	69.03°

5. Conclusions

In this work, the coupled origami is introduced by integrating two typical patterns belonging to rigid and non-rigid origami, opening new avenues for origami pattern innovation with enriched mechanical properties. Specifically, this coupled origami inherits the multistability of Leaf-like/Waterbomb origami, the foldability of Kresling origami, and the tunable stiffness enabled by these types. In addition, it also exhibits opposite chirality and enhanced kinematic DOFs, including two rotational modes, one axial telescopic mode, and one coupled compression-torsion mode that is associated with its folding mechanism. These features enable high flexibility and extensibility, making the coupled origami well-suited for soft continuum robotic systems that demand high compliance, versatility, and intuitive control.

Building on this flexibility, the origami-inspired manipulative robot is developed. By modeling each origami module as a parallel mechanism and establishing its kinematics model, we analyze the reachable workspace and diverse motion trajectories of the COISCMR-1, which demonstrate its capability to perform complex tasks with high dexterity. Moreover, under a straightforward tendon-driven actuation strategy, the COISCMR-1 achieves versatile bending motions in various degrees and omnidirectional manners.

Meanwhile, to further demonstrate the application potential of the independent scalability of the coupled origami, an origami-inspired mobile robot is constructed. Its configuration and principle are discussed, and its performance is evaluated through a continuum kinematic model that guides its underlying movements, including crawling, climbing, and turning. In addition, its ability to move continuously in complex environments is also demonstrated, highlighting its adaptability and potential for unstructured terrain navigation.

In summary, we construct a novel origami pattern and systematically analyze its mechanical properties, demonstrating its promising potential in robotic systems. This work enriches origami structures and opens new pathways for their functionalities into intelligent and adaptive machines. In future work, we are committed to studying this coupled origami and its mechanical behaviors, and expanding its applications to wider fields. In addition, we aim to further enhance the functionality and performance of both origami-inspired soft continuum robots, advancing them toward greater intelligence, environmental adaptability, and task-specific capabilities.

Experimental section / methods

Kinematic Analysis. The analysis model that considers kinematic features of creases and panels is challenging, as it involves both complex crease patterns and material properties. Therefore, using equivalent parallel mechanisms or continuum mechanisms to analyze the mechanical properties of origami/origami-inspired structures is feasible and effective. These two approaches are essentially identical, with the former being used to analyze the workspace of the coupled origami and the COISCMR-1, while the latter is used to guide the motion of the COISCMR-2.

Statistical Analysis. To characterize the structural performance of the coupled origami, we conduct five tests for each state using a sufficient number of samples. While manual manufacturing and testing methods bring certain challenges to data accuracy, the standardized fabrication process and the assistance of 3D modeling software ensure the reliability of these reference data shown in Fig. 2. For the origami-inspired mobile robot, its tests are carried out under varying tendon extensions guided by the kinematic model, with at least five trials for each data point. Although the mobility is affected by the material flexibility, friction effects, and even fluctuations in the power supply, which are reflected in the test data, the errors remain within an acceptable range. The mean and standard deviation of the measurements are used to evaluate their performance.

Sample Fabrication. The samples of the coupled origami and the COISCMR-1 are made of 180 g/m^2 cardboard, while the COISCMR-2 is made of 0.5 mm-thick polypropylene material. They are patterned using a laser cutter and then folded by hand along the mountain/valley creases. In addition, 3D printing is utilized to produce most of the remaining parts, excluding standard components. Detailed information is shown in *SI Note* 12 and Figs. S15-S18.

CRediT authorship contribution statement

Xiaolei Wang: Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization. **Yichen Wang:** Validation, Investigation. **Haibo Qu:** Writing – review & editing, Funding acquisition, Conceptualization. **Haoqian Wang:** Software, Data curation. **Wenju Liu:** Visualization, Resources, Data curation. **Jiaqiang Yao:** Software, Formal analysis. **Zhizhen Zhou:** Resources, Formal analysis, Data curation. **Sheng Guo:** Writing – review & editing, Funding acquisition. **Jinkyu Yang:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ijmecsci.2025.110620.

Data availability

Data will be made available on request.

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